ACVSD Lab2

電機四 B11901063 李佑倫

1.

(1) 8 paths

(2)

```
Operating Conditions: ff0p88v125c   Library: N16ADFP_StdCellff0p88v125c_ccs
Wire Load Model Mode: segmented

  Startpoint: FF1_reg (rising edge-triggered flip-flop clocked by CLKA)
  Endpoint: FF2_reg (rising edge-triggered flip-flop clocked by CLKA)
  Path Group: CLKA
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  CDC                ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs

  Point                                    Incr       Path
  -----------------------------------------------------------
  clock CLKA (rise edge)                   0.00       0.00
  clock network delay (ideal)              0.00       0.00
  FF1_reg/CP (DFQD2BWP16P90LVT)            0.00       0.00 r
  FF1_reg/Q (DFQD2BWP16P90LVT) <-          0.02       0.02 r
  FF2_reg/D (DFQD2BWP16P90LVT)             0.00       0.02 r
  data arrival time                                   0.02

  clock CLKA (rise edge)                  10.00      10.00
  clock network delay (ideal)              0.00      10.00
  FF2_reg/CP (DFQD2BWP16P90LVT)            0.00      10.00 r
  library setup time                      -0.01       9.99
  data required time                                  9.99
  -----------------------------------------------------------
  data required time                                  9.99
  data arrival time                                  -0.02
  -----------------------------------------------------------
  slack (MET)                                        9.97


  Startpoint: FF1_reg (rising edge-triggered flip-flop clocked by CLKB)
  Endpoint: FF2_reg (rising edge-triggered flip-flop clocked by CLKB)
  Path Group: CLKB
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  CDC                ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs

  Point                                    Incr       Path
  -----------------------------------------------------------
  clock CLKB (rise edge)                   0.00       0.00
  clock network delay (ideal)              0.00       0.00
  FF1_reg/CP (DFQD2BWP16P90LVT)            0.00       0.00 r
  FF1_reg/Q (DFQD2BWP16P90LVT) <-          0.02       0.02 r
  FF2_reg/D (DFQD2BWP16P90LVT)             0.00       0.02 r
  data arrival time                                   0.02

  clock CLKB (rise edge)                  20.00      20.00
  clock network delay (ideal)              0.00      20.00
  FF2_reg/CP (DFQD2BWP16P90LVT)            0.00      20.00 r
  library setup time                      -0.01      19.99
  data required time                                 19.99
  -----------------------------------------------------------
  data required time                                 19.99
  data arrival time                                  -0.02
  -----------------------------------------------------------
  slack (MET)                                        19.97
```

(3)

```
Operating Conditions: ff0p88v125c   Library: N16ADFP_StdCellff0p88v125c_ccs
Wire Load Model Mode: segmented

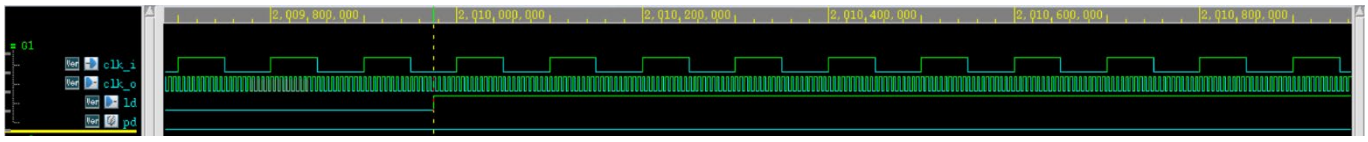
  Startpoint: FF2_reg (rising edge-triggered flip-flop clocked by CLKB)
  Endpoint: FF3_reg (rising edge-triggered flip-flop clocked by CLKC)
  Path Group: (none)
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  CDC                ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs

  Point                                    Incr       Path
  -----------------------------------------------------------
  FF2_reg/CP (DFQD2BWP16P90LVT)            0.00       0.00 r
  FF2_reg/Q (DFQD2BWP16P90LVT) <-          0.03       0.03 f
  FF3_reg/D (DFQD2BWP16P90LVT)             0.00       0.03 f
  data arrival time                                   0.03
  -----------------------------------------------------------
  (Path is unconstrained)
```

2.

(1)



(2)

```
*[PLL-INFO] PLL is initialized. Current time                1000 ns
*[PLL-INFO] PLL leaves power-down mode! at time 1010 ns
*[PLL-INFO] PLL acquires lock after 199998 ns!, Now 200998 ns
*[PLL-INFO] FOUT   =2000.000000MHZ
*[PLL-INFO] FIN    =100.000000MHz
*[PLL-INFO] R[4:0] =00011
*[PLL-INFO] F[7:0] =00100111
*[PLL-INFO] OD[2:0]=000
```